

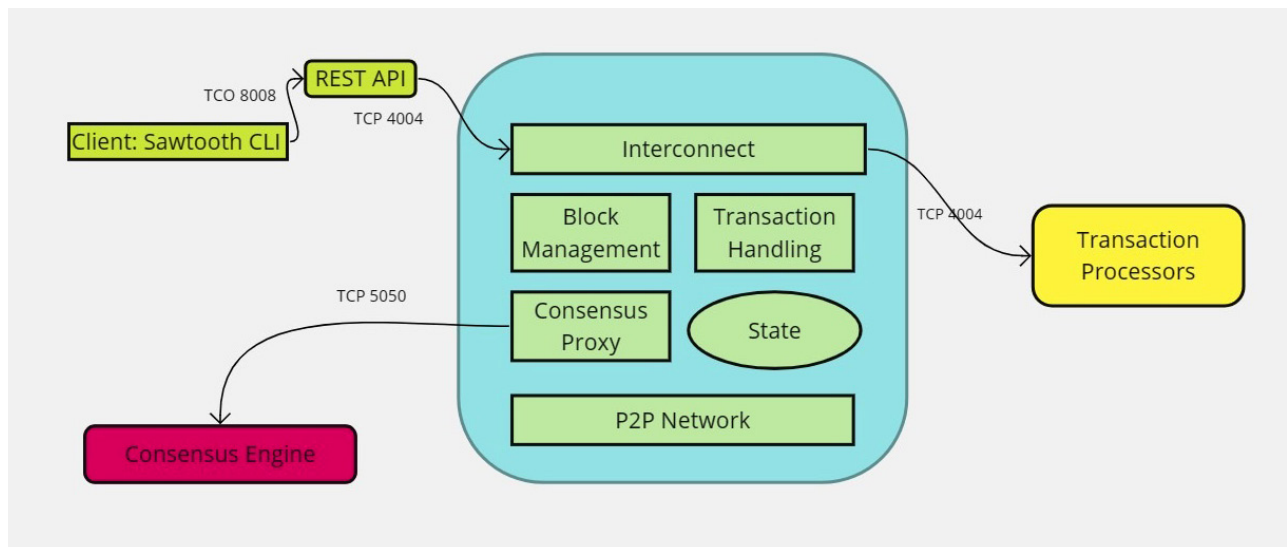


Figure 2: Sawtooth validator

outputs (addresses in the state that the transaction can read from or write to).

- **Header signature:** This digital signature is created by signing the header with the transaction creator's private key. It ensures the integrity and authenticity of the transaction.
- **Payload:** This is the actual data of the transaction, which is interpreted by the transaction processor. The structure of the payload is determined by the specific transaction family's requirements. Transactions are submitted to the

Sawtooth network through clients, which construct the transaction payload, wrap it in a transaction envelope (including the header and signature), and send it to the network.

Batches

A batch is a collection of transactions submitted together to the network. Batches allow the grouping of related transactions so that they are all either committed to the state or rejected, ensuring atomicity. Each batch includes:

- **Transactions list:** An ordered list of transactions that belong to the batch. The transactions are executed in the order they appear in the list.

- **Header:** Contains metadata about the batch, similar to transactions. It includes the signer's public key and a list of transaction identifiers (signatures) in the batch.
- **Header signature:** This digital signature of the batch header is signed with the batch creator's private key. It ensures the integrity and authenticity of the batch.

BatchHeader

The BatchHeader contains metadata about the batch, including information about the transactions contained within it.

- **string signer_public_key:** This public key of the entity that signed the batch is used to verify the header_signature.
- **Repeated string transaction_ids:** A list of identifiers for each transaction included in the batch. This ensures that the transactions can be uniquely identified and referenced.

Transaction

A transaction represents an individual operation or action that is intended to be executed against the ledger. Each transaction within a batch is executed in the order it is listed.

- **Bytes header:** A binary representation of the TransactionHeader, it encapsulates various details about the transaction.
- **string header_signature:** A digital signature of the TransactionHeader, serving a similar purpose as the header_signature in the BatchHeader but for individual transactions.
- **bytes payload:** The actual data or instructions that the transaction is carrying. This is what gets executed against the ledger.

TransactionHeader

The TransactionHeader provides the necessary context and metadata for processing the transaction.

- **string batcher_public_key:** The public key of the entity responsible for batching this transaction. This might be the same as the signer_public_key if the transaction's creator also created the batch.
- **repeated string dependencies:** Identifiers of other transactions that must be executed before this one.
- **string family_name** and **string family_version:** Identifiers for the transaction processor that can execute this transaction, allowing